ASCII MASTER FLOW — PANEL 1/12: Fission-to-grid chain

FISSION -> heavy nucleus splits and releases heat plus neutrons
COOLANT -> removes core heat
STEAM SYSTEM -> converts heat into working-fluid energy
TURBINE-GENERATOR -> produces electricity
RULE -> reactor capacity is not electricity generated

ASCII MASTER FLOW — PANEL 2/12: Criticality milestone ladder

```
SUBCRITICAL -> chain reaction declines
FIRST CRITICALITY -> controlled self-sustaining reaction begins
POWER ASCENSION -> tested increase toward rated conditions
GRID SYNCHRONISATION -> generator connects to grid
COMMERCIAL OPERATION -> later declared operating status
```

ASCII MASTER FLOW — PANEL 3/12: Core-control function map

```
CONTROL RODS -> absorb neutrons and regulate reactivity
MODERATOR -> slows neutrons in a thermal reactor
COOLANT -> transfers heat out of the core
CONTAINMENT -> barrier against radioactive release
RULE -> four functions, four different answers
```

ASCII MASTER FLOW — PANEL 4/12: Thermal-fast comparison

THERMAL REACTOR -> moderated slower-neutron spectrum
FAST REACTOR -> no moderator and fast-neutron spectrum
PHWR -> heavy-water moderated Stage-1 workhorse
FBR -> breeding objective with fertile blanket
RULE -> coolant alone does not define the spectrum

ASCII MASTER FLOW — PANEL 5/12: Fissile-fertile conversion map

```
U-235 / PU-239 / U-233 -> fissile materials
U-238 -> neutron capture -> PU-239 pathway
TH-232 -> neutron capture and decay -> U-233 pathway
FERTILE MATERIAL -> needs irradiation and conversion
RULE -> resource abundance is not ready reactor fuel
```

ASCII MASTER FLOW — PANEL 6/12: Three-stage programme rail

STAGE 1 -> PHWR, natural uranium and plutonium-bearing spent fuel
REPROCESSING -> separates material for the breeder bridge
STAGE 2 -> fast breeder and fissile-inventory multiplication
THORIUM IRRADIATION -> prepares U-233 route
STAGE 3 -> long-term thorium-U-233 utilisation horizon

ASCII MASTER FLOW — PANEL 7/12: PFBR status card

DESIGN -> IGCAR fast-reactor and sodium expertise
BUILD / COMMISSION -> BHAVINI project responsibility
RATING -> 500 MWe design value in fetched DAE text
6 APRIL 2026 -> first criticality achieved
NOT ESTABLISHED -> grid synchronisation or commercial operation

ASCII MASTER FLOW — PANEL 8/12: Closed fuel-cycle loop

```
REACTOR IRRADIATION -> changes isotopic composition
COOLING / HANDLING -> manages heat and radiation
REPROCESSING -> separates reusable fissile material
REMOTE FABRICATION -> prepares new fuel
REUSE / WASTE -> closes material accounting loop
```

ASCII MASTER FLOW — PANEL 9/12: Institutional separation

```
DAE -> governmental policy and strategic umbrella
BARC -> reactor and fuel-cycle R&D
NPCIL -> commercial civilian plant developer and operator
AERB -> safety review and regulatory oversight
RULE -> promoter, laboratory, operator and regulator differ
```

ASCII MASTER FLOW — PANEL 10/12: Fuel-and-project chain

```
UCIL -> uranium mining and milling
NFC -> fuel and reactor-component fabrication
IREL -> monazite, thorium and rare-earth processing link
IGCAR -> fast-reactor design and sodium R&D
BHAVINI -> PFBR project execution and operation
MUST REMEMBER: Fission splits heavy nuclei and is controlled through fuel, moderator,...
```

ASCII MASTER FLOW — PANEL 11/12: Reactor taxonomy

PHWR -> heavy water and natural-uranium Stage-1 role
PWR / BWR -> light-water systems with different steam cycles
FBR -> fast spectrum and breeding objective
AHWR / SMALL-REACTOR LINES -> design or development status
RULE -> type and deployment status must be stated together
CLOSE DISTINCTION: Fissile is not fertile, enrichment is not reprocessing, heavy water is...

ASCII MASTER FLOW — PANEL 12/12: Nuclear answer spine

```
DEFINE -> fission, criticality, fissile, fertile and fuel cycle
TRACE -> Stage 1, plutonium bridge, Stage 2 and thorium horizon
MAP -> DAE BARC NPCIL AERB IGCAR BHAVINI and fuel chain
SEPARATE -> capacity criticality grid generation and legal status
VERIFY -> dated fleet fuel reactor law cost and schedule evidence
EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Trace PHWR uranium, fast-breeder...
```